JOSHUA LE

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SUMMARY

Software engineer specializing in real-time simulation and autonomous systems, with experience spanning defense training simulators, digital twin platforms, and NASA-funded autonomous robotics research. Proficient in C++, C, C#, and Python, with hands-on work building physics-based 3D simulations, developing autonomous rover and drone platforms, and processing LiDAR sensor data in ROS on Linux, delivered within cross-disciplinary engineering teams.

EXPERIENCE

Software Engineer — CACI International, San Antonio, Texas **March 2024 – Present**

- Design, implement, and maintain core features of a real-time medical triage training simulator used to train medical personnel and support research studies, improving simulation performance by 50%.
- Engineered redundant data-persistence methods for logged research telemetry, ensuring 100% of experiment data is captured and saved.
- Built and improved automated testing workflows for React and MongoDB-based simulation systems within an Agile development process.

Software Engineer Intern — CACI International, San Antonio, Texas **March 2022 – March 2024**

- Led development of a real-time dashboard for monitoring network security events, integrating multiple data sources in React to support rapid incident response.
- Developed an interactive digital twin simulation platform in C# and Unity 3D, streamlining penetration testing and security evaluation workflows.
- Built a photogrammetry-based digital twin in Unity 3D for a STEM-education partnership with Port San Antonio.

Graduate Researcher — UTSA SAVE Lab, San Antonio, Texas **June 2022 – December 2023**

- Conducted VR cybersickness and cognitive-load research contributing to a predictive model with over 90% accuracy; results published in an IEEE paper (2025).
- Created a Unity 3D augmented-reality application integrating real-time geospatial data with realistic plume and fire hazard simulations to improve situational awareness for emergency response teams.

Undergraduate Researcher — NASA MIRO CAMEE, San Antonio, Texas **November 2020 – December 2022**

- Developed a fully autonomous robotic platform pairing a 3D-printed rocker-bogie rover with an autonomous drone for LiDAR terrain mapping and data acquisition.
- Processed and cleaned LiDAR point-cloud sensor data using ROS (C++ and Python), improving data quality for coastal reconnaissance mapping.

LANGUAGES & TECHNOLOGIES

Languages: C++, C, C#, Python, JavaScript, TypeScript

Simulation & Robotics: Unity 3D, ROS, LiDAR point-cloud processing, autonomous robotic systems, digital twins, AR/VR, photogrammetry, MATLAB, Blender, 3ds Max

Tools & Platforms: Git, Docker, Linux, React, MongoDB, MySQL

EDUCATION

B.S. Computer Science — University of Texas at San Antonio **May 2023**

Minor in Cybersecurity and Digital Forensics • GPA 3.5 • Honors Student

PUBLICATIONS

- “Predicting and Explaining Cognitive Load, Attention, and Working Memory in Virtual Multitasking” — IEEE, 2025.
- “Mazed and Confused: A Dataset of Cybersickness, Working Memory, Mental Load, Physical Load, and Attention During a Real Walking Task in VR” — 2024.

LEADERSHIP & OUTREACH

Mentor, FIRST Tech Challenge — Family Service Association, San Antonio **July 2021 – Present**

- Mentor a 5-student robotics team in engineering design and programming to consistent regional-qualification finishes; helped secure and manage $10,000 in equipment to expand STEM access in under-served communities.